Lecture 9.

Solution of Sessional 1.

Q1:- a). $\forall x P(x)$. $P(x) = x \cap B \neq \Phi$. $x \{\{a,c\},\{e\},\{d\}\}$.

$B = \{a,b,c,d\}$.

$= P(\{a,c\}) \wedge P(\{e\}) \wedge P(\{d\})$.

$= \{a,c\} \cap \{a,b,c,d\} \neq \Phi \wedge \{e\} \cap \{a,b,c,d\} \neq \Phi \wedge$.

$\{d\} \cap \{a,b,c,d\} \neq \Phi$.

$\{a\} \neq \Phi \wedge \Phi \neq \Phi \wedge \{d\} \neq \Phi$.

$T \wedge F \wedge T = F$.

b) $\exists x\, p(x,y)$ $x \in \{0,1,2\}$ $y \in \{1\}$.

$P(x,y) = x+y = 1$.

$= P(0,1) \vee P(1,1) \vee P(2,1)$.

$= 0+1=1 \vee 1+1=1 \vee 2+1=1$

$T \vee F \vee F = T$.

Q2:- $\neg \boxed{S} \vee \boxed{\neg L}$. (Contrapositive). $p \to q = \neg p \vee q$.

$p \quad q$.

$\neg S = R \wedge W$.

$L = \neg U$.

$= S \to \neg L$. (Contrapositive).

a) Converse $\neg S \to L = S \vee L = \neg(R \wedge W) \vee \neg U$

$= \neg R \vee \neg W \vee \neg U$

b) Inverse $\neg L \to S = L \vee S = \neg U \vee \neg R \vee \neg W$.

c) Implication:- $L \to \neg S$

$\neg U \to R \wedge W. = \neg(\neg U) \vee (R \wedge W)$

$U \vee (R \wedge W)$.

Q3:-

$$\boxed{\begin{array}{llll} p = \text{Sam is Verifier} & & \neg p = \text{Sam is disruptor} \\ q = \text{Riley " "} & & \neg q = \text{Riley " "} \\ r = \text{Noor " "} & & \neg r = \text{Noor " "} \end{array}}$$

$$\boxed{\begin{array}{l} (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \\ p \rightarrow \neg r \\ \neg q \end{array}}$$

CASE 1:-

$$(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) = T$$

$$p \rightarrow \neg r = T$$

$$\neg q = T$$

$$\boxed{\begin{array}{l} p = T \\ q = T \\ r = T \end{array}}$$

⋮

CASE 8:-

Q4:-

$P \longleftrightarrow Q = T$ — (1) ✓

$Q \rightarrow R = T$ — (2) ✓

$R \longleftrightarrow S = T$ — (3) ✓

$\neg S \vee P = T$ — (4) ✓

$\neg P = T$ — (5) ✓

R	S	$R \leftrightarrow S$
~~T~~	~~T~~	~~T~~
[illegible]	~~F~~	~~F~~
[illegible]	~~T~~	~~F~~
F	F	T

from (5) $P = F$ — (6) ✓

put (6) in (4) $S = F$ — (7) ✓

put (7) in (3) $R = F$ — (8) ✓

" (8) in (2) $Q = F$ — (9) ✓

$P = F$ & $Q = F$ in (1)

$F \longleftrightarrow F = T$

✓ Consistent.

Q	R	$Q \rightarrow R$
~~T~~	~~T~~	~~T~~
~~T~~	~~F~~	~~F~~
~~F~~	~~T~~	~~T~~
F	F	T